

NBA Player Performance and Salary Analysis

An Analytical Study of the 2025–26 Season

Research Question

How do NBA player performance statistics relate to player salary during the 2025–26 NBA season?

Prepared by

Masego & Matthias

MA346 — Data Science

Project 2

2025–26 NBA Season

Executive Summary

This report examines how player performance statistics relate to player salaries during the 2025–26 NBA season. The analysis combines two publicly available datasets — individual performance statistics and current-season salary information — and applies a sequence of statistical methods to quantify the relationship and assess how reliable that relationship is across different parts of the salary distribution.

The headline finding is that on-court performance is strongly associated with compensation, but does not fully explain it. Points per game shows the strongest single relationship with salary, with a correlation of approximately 0.75. A simple linear model using points alone explains roughly 56% of the variation in salary, and a broader multivariate model using points, assists, rebounds, and minutes played explains roughly 58%. The remaining variation is attributable to factors outside the dataset — among them player reputation, contract timing, experience, injuries, team needs, and salary-cap dynamics.

Finding	Result
Strongest correlate of salary	Points per game ($r \approx 0.75$)
Second-strongest correlate	Minutes played ($r \approx 0.60$)
Variation in salary explained by points alone (R^2)	$\approx 56.2\%$
Variation explained by full multivariate model (R^2)	$\approx 58.3\%$
Statistical significance of overall model	F-test p-value < 0.001
Estimated change in salary per +1 point (simple model)	$\approx +\$1.55$ million
Estimated change in salary per +1 point (median model)	$\approx +\$1.94$ million

Table 1. Summary of headline statistical findings.

Project Goal

The goal of this project is to evaluate how NBA player performance statistics relate to player salary during the 2025–26 NBA season. Specifically, the analysis seeks to determine whether higher-performing players tend to earn higher salaries and to identify which performance metrics are most closely associated with compensation.

To answer this question, NBA player performance data was merged with NBA player salary data into a single, player-level dataset. This combined dataset enables direct comparison between each player's salary and a set of on-court performance measures: points, assists, rebounds, and minutes played per game.

Data Sources and Preparation

Performance statistics alone would not allow for an analysis of compensation, and salary data alone would not show how players actually performed on the court. Combining the two sources produced a dataset in which each row represents a single player during the 2025–26 NBA season with both performance and salary information attached.

Performance Data

Player performance data was sourced from Stathead Basketball, part of the Sports Reference basketball database. Stathead provides structured, season-level NBA statistics covering points, assists, rebounds, and minutes played, among other variables. These four metrics were chosen because they represent widely accepted measures of individual production: points capture scoring output, assists capture playmaking, rebounds capture possession impact, and minutes played capture how much each player is being used by his team.

Salary Data

Salary information was sourced from Basketball-Reference and reflects each player's compensation for the 2025–26 NBA season. This dataset is essential to the project because salary is the outcome variable of interest. Pairing salary with performance enables a direct examination of whether stronger on-court production tends to coincide with higher pay.

Why These Datasets Fit the Project Goal

Together, the two sources cover both sides of the research question. Performance metrics describe what each player contributes during games, while salary captures how the league compensates that contribution. Both datasets are organized at the player level for the same season, which makes them naturally compatible for joint analysis once aligned on player and season identifiers.

Data Cleaning

Several preparation steps were required before the two datasets could be combined. Column headers were standardized because the original files did not load with usable headers. Unnecessary columns were removed so the analysis focused on the variables relevant to the research question — player name, season, points, assists, rebounds, minutes played, and salary. A common season identifier ("2025–26") was added to the salary file so that both datasets shared the same key fields.

Player names were also normalized by stripping leading and trailing whitespace. This was important because hidden whitespace causes name strings to be treated as different values during a join, even though they look identical to a human reader. Duplicate player-season rows were removed from each dataset before merging so that each player appeared only once per season per source.

Merging Process and Data Quality Checks

How the Datasets Were Merged

The two datasets were combined using player name and season as matching keys. The result is a single table in which each row represents a player who appeared in both the performance data and the salary data for the 2025–26 season.

Why an Inner Join Was Appropriate

An inner join was used because the analysis requires both pieces of information for every player. A player who appeared only in the performance dataset, with no salary on file, could not be used to study compensation. Conversely, a player who appeared only in the salary dataset, with no performance statistics, could not be used to study performance. The inner join produced the cleanest possible dataset for the project's research question by keeping only those players for whom both sides of the equation were available.

Final Variable Set

After merging, the dataset contained the following key variables:

Variable	Meaning	Role in the Project
Player	NBA player name	Used to match the two datasets
Season	NBA season	Confirms data refers to the same season
PTS	Points per game	Measures scoring performance
AST	Assists per game	Measures playmaking
TRB	Total rebounds per game	Measures rebounding production
MP	Minutes played per game	Measures player usage
Salary	Player salary (USD)	Main outcome variable

Table 2. Variables retained in the final merged dataset and their role in the analysis.

Sample of the Merged Dataset

A representative sample of ten rows from the merged dataset is shown below. Each row pairs a player's per-game performance statistics with that player's 2025–26 salary.

Player	Season	PTS	AST	TRB	MP	Salary 2025–26
Amen Thompson	2025–26	18.3	5.3	7.8	37.4	\$9,690,600
Robert Williams	2025–26	6.7	1.0	7.0	17.1	\$13,285,713
Malaki Branham	2025–26	4.6	0.8	1.6	9.8	\$4,962,033
Pacôme Dadiet	2025–26	1.7	0.4	0.9	4.7	\$2,847,600
Amir Coffey	2025–26	3.2	0.6	1.2	10.7	\$2,296,274
Taurean Prince	2025–26	9.2	1.8	3.1	23.5	\$3,303,774
E.J. Liddell	2025–26	5.7	0.9	2.7	13.4	\$706,898
JD Davison	2025–26	2.5	1.3	1.2	7.8	\$2,270,735

Player	Season	PTS	AST	TRB	MP	Salary 2025-26
Anfernee Simons	2025-26	14.3	2.4	2.5	24.9	\$27,678,571
Tyler Smith	2025-26	4.7	0.4	2.8	13.8	\$1,955,377

Table 3. Random sample of ten players from the merged dataset, used as a manual quality check.

Merge Quality Checks

Several validation checks were run on the merged dataset. Row counts before and after the merge were compared, the presence of columns from both source files was confirmed, and the uniqueness of player-season rows was verified. These checks matter because a poorly executed merge can introduce duplicate players, missing values, or mismatched records that quietly distort downstream analysis.

Missing-value checks identified one player with a missing salary; this row was removed because salary is the project's outcome variable and the row could not contribute to either the regression or the correlation analysis. Finally, a random sample of the merged dataset was reviewed manually to confirm that player names, performance statistics, and salary values lined up correctly. After these checks, the cleaned dataset was exported to CSV and used as the foundation for all subsequent analysis.

Exploratory Analysis

Initial analysis used correlations and scatterplots to characterize the relationships between salary and the four performance metrics. Three observations stood out:

- Salary is most strongly correlated with **points ($r \approx 0.75$)**, indicating a strong positive linear association.
- Minutes played also shows a strong positive relationship ($r \approx 0.60$).
- Assists and rebounds show moderate positive correlations ($r \approx 0.57$ and 0.45 , respectively).

Note on terminology: a correlation coefficient ranges from -1 to $+1$. Values near $+1$ indicate a strong positive linear association; values near 0 indicate little linear association.

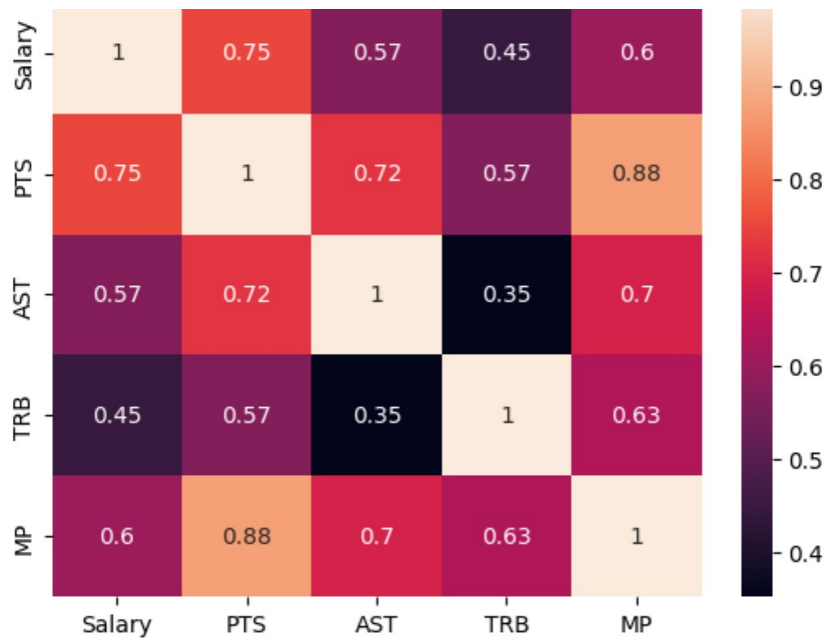


Figure 1. Correlation matrix showing that points has the strongest relationship with salary, followed by minutes played.

The scatterplots of salary against points and salary against minutes played both show clear upward trends. The spread of the data also widens at higher values of each performance metric, meaning that high-performing players show a wider range of salaries than low-performing players. This suggests that while performance is strongly associated with salary on average, the relationship becomes less consistent at the top end of the distribution.



Figure 2. Scatterplots of salary against minutes played (top) and points per game (bottom), showing positive relationships and increasing variability at higher values.

Statistical Modeling Results

Three modeling approaches were used to quantify the relationships and to test how stable they are across the salary distribution: simple linear regression, multivariate linear regression (with an analysis-of-variance breakdown of feature contributions), and median (quantile) regression.

Baseline Model: Salary Predicted by Points

A simple linear regression model using points per game as the only predictor establishes a strong baseline.

- $R^2 = 0.562$ — points alone explains roughly 56% of the variation in salary.
- **Estimated coefficient on points $\approx$ \$1.55 million per additional point per game**, and is highly statistically significant ($p < 0.001$).

Plain-English read: scoring is by far the most predictive single feature, but it does not capture the full picture of how salaries are determined. R^2 (the "coefficient of determination") expresses the share of variation in the outcome — here, salary — that the model accounts for. A value of 0.56 means a little over half of that variation is captured by points alone.

Multivariate Model: Adding Assists, Rebounds, and Minutes

A multiple linear regression was then estimated using all four performance metrics — points, assists, rebounds, and minutes played — as predictors. Multiple regression measures how each variable contributes to the outcome while holding the others constant.

Model Performance

- $R^2 = 0.583$, a modest improvement over the points-only model (about two percentage points).
- The model is highly statistically significant overall (F-test p-value < 0.001).

Feature Contributions

- Points remains the strongest predictor of salary.
- Assists and rebounds are statistically significant and add incremental predictive value.
- Minutes played has a negative coefficient in the joint model. This is most likely a sign of multicollinearity — the predictors are themselves highly correlated, which means each variable's individual coefficient becomes harder to interpret in isolation.

Plain-English read: adding more performance metrics improves the model, but only marginally. The very high correlation between points and minutes played ($r \approx 0.88$) means the two variables carry overlapping information. When highly redundant variables are entered into the same model, individual coefficients can flip sign or behave counter-intuitively, even though the overall model remains valid for prediction.

Feature Significance (ANOVA)

An ANOVA (analysis of variance) breakdown confirms that all four performance variables make statistically significant contributions to the model. ANOVA tests whether each variable explains a meaningful share of variation beyond what the other variables already explain.

- Points is the dominant feature.
- Assists and rebounds add incremental predictive power.
- Minutes played remains statistically significant despite its negative coefficient in the joint model.

This pattern reinforces that salary is associated with multiple aspects of performance — not scoring alone.

Distribution-Sensitive Modeling: Quantile Regression

Standard linear regression estimates effects at the average of the salary distribution. Because NBA salaries are highly skewed, with a small number of players earning many times the median, the average can be a misleading reference point. To check whether the conclusions hold for a typical player rather than just the league-wide average, a quantile regression was estimated at the median (the 50th percentile).

Key Insights

- **Estimated coefficient on points $\approx$ \$1.94 million per additional point at the median**, consistent with — and slightly stronger than — the OLS estimate.
- Assists and rebounds remain statistically significant and slightly stronger at the median.
- Minutes played remains negative and significant in the joint model, mirroring the OLS result.

Plain-English read: the relationship between performance and salary is not uniform across the salary distribution. Quantile regression, by modeling the median rather than the mean, is also far less sensitive to extreme outliers — for example, a handful of supermax contracts. The fact that points remains a strong predictor at the median indicates that the headline finding is not driven by a few highly paid stars.

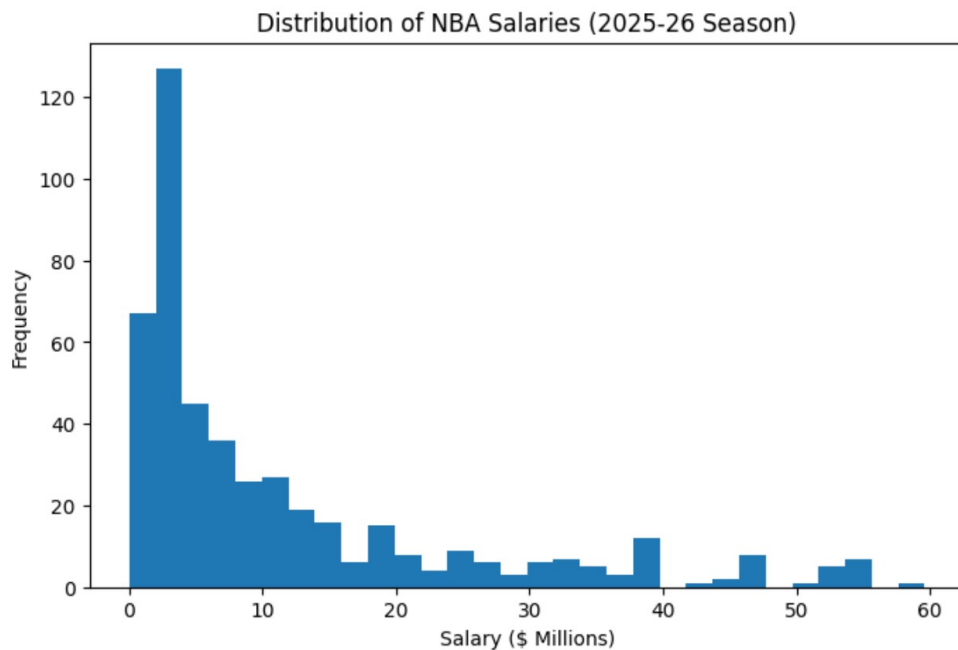


Figure 3. Salary distribution showing that most NBA players earn below \$10 million, with a smaller group of highly paid players forming a long right tail.

Players were also grouped into low, medium, and high scoring tiers and their salary distributions compared. High-scoring players have higher medians and substantially higher maximum salaries than low- and medium-scoring players, consistent with the regression results.

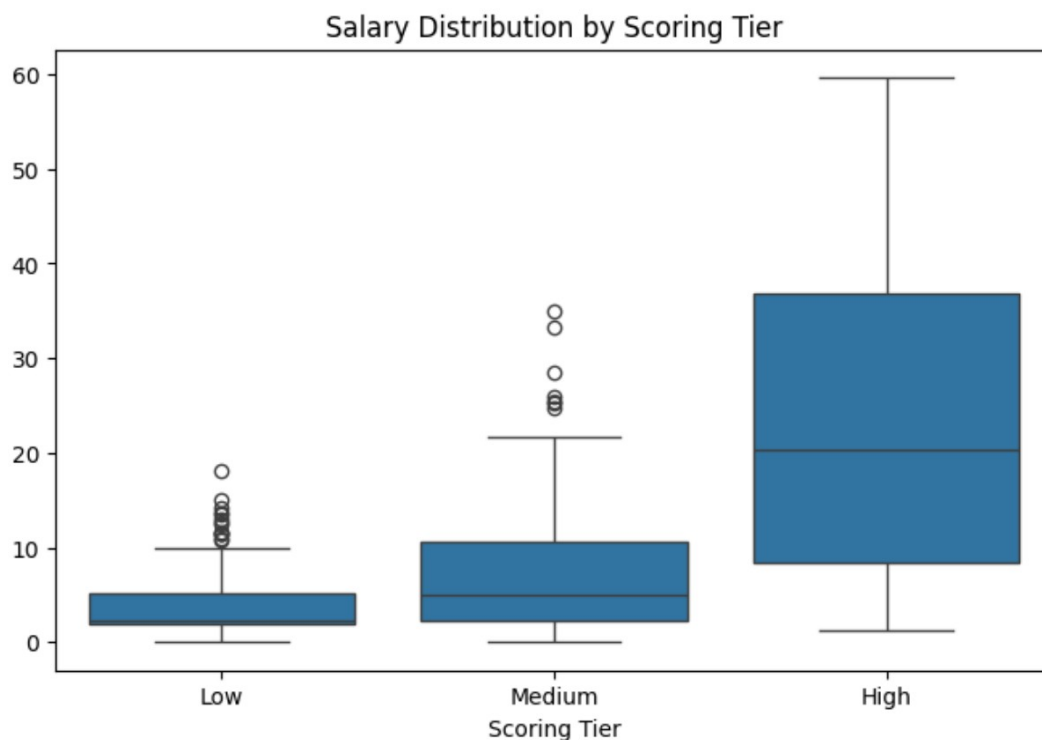


Figure 4. Salary distribution by scoring tier. High-scoring players show higher medians and higher upper-tier salaries than medium- and low-scoring players, with notable outliers in every tier.

The presence of outliers in every tier is itself informative. Some low- or medium-scoring players earn high salaries because of experience, reputation, defensive value, or the role they play within a team's system. Conversely, some high-scoring players earn relatively modest salaries because they are young, on rookie-scale contracts, or have not yet signed their first major long-term deal. These exceptions support the broader finding that scoring is strongly related to salary, but salary is also shaped by factors beyond current-season performance.

Model Diagnostics and Limitations

Residual Analysis

A residual is the gap between the salary the model predicts for a player and the salary that player actually earns. Plotting residuals against predicted salaries provides a visual check of whether the model's errors are well behaved.

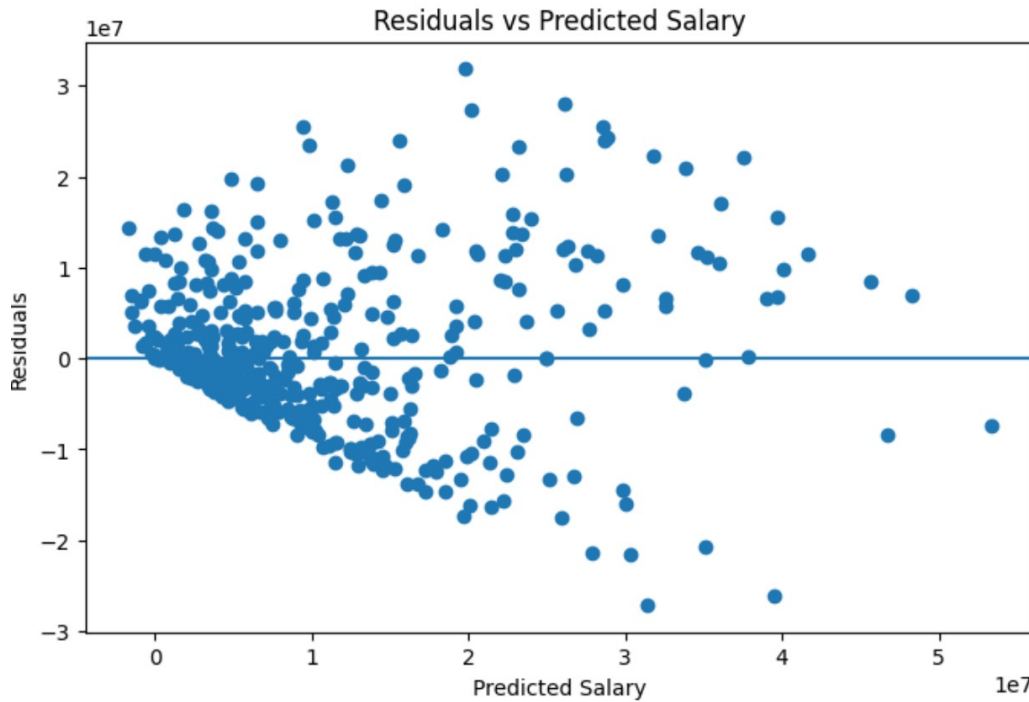


Figure 5. Residuals versus predicted salary. The fan-shaped pattern indicates that prediction error grows for higher-paid players — a sign of heteroskedasticity.

The residual plot shows a clear fan-shaped pattern: errors are tightly clustered near zero for lower predicted salaries and spread much more widely as predicted salary rises. In statistical terms, this is heteroskedasticity — non-constant variance — and it has two important implications:

- Prediction error grows as predicted salary increases.
- The model is therefore less reliable for top-tier players than for the league as a whole.

Implications and Possible Extensions

- Linear models likely cannot fully capture the dynamics that determine star-level salaries, where contract structure, brand value, and salary-cap rules play an outsized role.
- Modeling the logarithm of salary instead of raw salary, or using non-linear methods such as gradient boosting or quantile regression at multiple percentiles, could improve fit and stabilize errors at the top of the distribution.

Multicollinearity

The very high correlation between points and minutes played ($r \approx 0.88$) suggests redundancy between these features. Multicollinearity can inflate standard errors and make individual coefficients unstable, which is one likely explanation for the negative coefficient on minutes played in the joint model. The overall model remains valid as a predictor of salary, but coefficients on individual variables should be interpreted with this redundancy in mind.

Other Limitations

- The dataset covers a single season (2025–26). Salary decisions reflect multi-year contracts and historical performance that the current-season snapshot cannot represent.
- Defensive metrics, advanced statistics (e.g., player efficiency rating, win shares), and contract-status indicators are not included, even though they are likely relevant to compensation.
- The analysis is associational, not causal. Players are not randomly assigned to performance levels or salary tiers, so the results describe how the two move together — not how one causes the other.

Key Takeaways

- Performance metrics are strong predictors of NBA salary in the 2025–26 season.
- Points per game is the most informative single feature across every model tested — simple linear regression, multivariate regression, and median regression.
- Adding assists, rebounds, and minutes played improves explanatory power, but only marginally — about two percentage points of additional R^2 .
- The assumption of constant prediction error is violated: the model is most reliable for mid-salary players and least reliable for the highest-paid players.
- Comparing approaches matters. Median regression provides a robustness check that confirms the main finding holds for typical players, not just league averages affected by extreme outliers.

Interpretation in Context

Even with a statistically significant model, a meaningful share of salary variation remains unexplained. The data points to several factors that are plausibly important and not captured here:

- Player reputation and brand value
- Contract structure, length, and timing
- Years of experience and age
- Team-specific salary-cap strategy and roster needs
- Injury history and durability

The broader takeaway is that salary in the NBA is only partially explained by the kinds of observable, current-season performance metrics used here. A more complete predictive model would benefit from a richer feature set spanning advanced performance metrics, contract data, market dynamics, and team-context variables.

Conclusion

Across every modeling approach used in this analysis, NBA player performance is strongly associated with salary during the 2025–26 season. Points per game has the most consistent and largest relationship with compensation, and the regression models collectively show that performance metrics explain a meaningful — though incomplete — portion of the variation in salary.

At the same time, the analysis demonstrates clear limits to a performance-only view of pay. Diagnostic checks reveal that prediction error grows for higher-paid players, multicollinearity complicates the interpretation of individual coefficients, and a substantial share of salary variation remains unexplained even in the broader model. Salary decisions in the NBA are shaped by contract timing, experience, reputation, team needs, age, injury history, salary-cap rules, and broader market conditions — factors outside the scope of this dataset.

The overall conclusion, then, is best phrased in associational rather than causal terms: stronger performance generally corresponds with higher salary, but salary is most accurately understood as the joint outcome of measurable on-court production and a wider set of contractual and organizational factors.